

Robust Deep Video Watermarking Against Camera Recording Attacks via Temporal Distortion Modeling and Synchronization

Zhengdong Wang, Xin Liao*, Yufeng Wu

Abstract—Camera recording attacks pose a serious challenge to video watermarking due to the complex spatiotemporal distortions introduced during the playback–recording process. In particular, frame-rate mismatch causes temporal resampling and severe desynchronization, making embedded watermarks difficult to reliably decode. Existing deep video watermarking frameworks predominantly emphasize robustness against spatial or digital distortions but do not explicitly model the temporal distortions introduced by camera recording, resulting in substantial degradation in watermark extraction. To address this issue, we propose RecMark, a deep video watermarking framework that integrates differentiable spatiotemporal distortion modeling with learned frame synchronization. Specifically, we introduce a differentiable distortion layer that simulates spatial degradations and temporal desynchronization caused by mismatched playback and recording frame rates. A dedicated synchronization watermark is embedded in the frequency domain to provide frame-level synchronization cues and reconstruct temporally aligned frame groups, while the copyright watermark is embedded in the spatiotemporal domain through an encoder equipped with a Spatiotemporal Fusion Module (SFM). Extensive experiments demonstrate that RecMark achieves high extraction accuracy under camera recording attacks while preserving high perceptual video quality.

Index Terms—Video watermarking, camera recording attacks, temporal distortion.

I. INTRODUCTION

IN widespread video distribution scenarios, unauthorized screen recording using cameras poses a threat to copyright protection due to its ease of execution and difficulty of detection [1]. Video watermarking offers a promising solution by embedding imperceptible yet extractable information [2], [3], [4]. However, camera recording attacks involve capturing videos displayed on screens and re-digitizing them through cameras [5], introducing complex physical distortions beyond the scope of conventional digital-domain video watermarking. In contrast, they introduce a different degradation process characterized by coupled spatial–temporal distortions, including non-linear optical distortions induced by screen-camera interaction and frame rate mismatch [6], [7], [8]. As a result, the watermark signal suffers from both spatial degradation and temporal misalignment, leading to unreliable extraction.

This work is supported by New Generation Artificial Intelligence-National Science and Technology Major Project (2025ZD0123601), National Natural Science Foundation of China (Grant No. U22A2030), Hunan Provincial Funds for Distinguished Young Scholars (Grant No. 2024JJ2025), The science and technology innovation Program of Hunan Province (2025QK2008), Hunan Provincial Key Research and Development Program (Grant Nos. 2024AQ2027, 2025AQ2022).

Zhengdong Wang, Xin Liao, Yufeng Wu are with the College of Cyber Science and Technology, Hunan University, Changsha, China (e-mail: wangzdx@126.com, xinliao@hnu.edu.cn, wuyufeng0523@163.com).

*Corresponding author: Xin Liao

Copyright © 2026 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending an email to pubs-permissions@ieee.org.

This highlights the need for video watermarking methods specifically robust to camera recording attacks.

Traditional signal processing-based video watermarking methods have been explored for camera recording attacks but remain limited, as they are designed for specific noise types and struggle to generalize to compounded distortions [9], [10], [11], [12], [13]. Deep video watermarking methods leverage the powerful fitting capability of neural networks to address this problem [14], [15], [16]. DVMark [17] employs adversarial training with distortion layer to improve robustness against various spatial distortions. Nevertheless, existing deep approaches predominantly model spatial or digital distortions and do not explicitly account for the temporal distortions introduced by temporal resampling during the camera recording process.

In addition, existing deep learning-based methods typically segment videos into clips of fixed length and achieve group-wise watermark embedding due to the limitation of network. REVMARK [18] utilizes 3D-CNN to embed a watermark into fixed group of 8 frames. VideoSeal [19] embeds watermark at an interval of k frames. While these methods perform well against digital distortions, they are generally ineffective against camera recording attacks. This is because their fixed, group-based periodic embedding lacks temporal synchronization under severe temporal misalignment.

To address the limitations identified above, we propose RecMark, a deep video watermarking framework that integrates differentiable spatiotemporal distortion modeling of the camera recording process with learned frame synchronization. Unlike existing synchronization methods that rely on predefined features [10], [12], RecMark learns frame-level synchronization cues from distorted videos for copyright watermark decoding. Specifically, we introduce a differentiable distortion layer that simulates spatial degradations and temporal desynchronization caused by mismatched playback and recording frame rates. A dedicated synchronization watermark is embedded in the frequency domain to facilitate frame alignment, while the copyright watermark is embedded in the spatiotemporal domain through an encoder equipped with a Spatiotemporal Fusion Module (SFM), enhancing redundancy and consistency with dynamic video content.

In summary, we outline our main contributions as follows:

- To address the lack of effective temporal distortion modeling in existing deep video watermarking methods, we introduce a differentiable spatiotemporal distortion layer that simulates camera recording-induced spatial degradations and temporal desynchronization, including under-sampling, over-sampling, and mixed-sampling caused by frame-rate mismatch. This enables robust representation

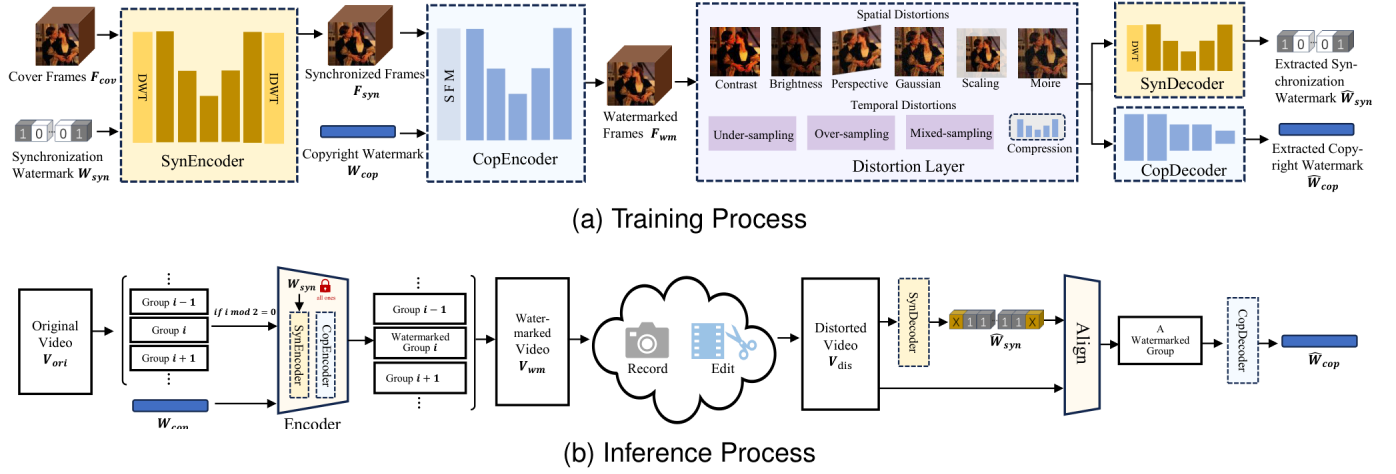


Fig. 1. In the training process, we randomly sample 8-frame cover frames F_{cov} , 8-bit synchronization watermark W_{syn} and 96-bit copyright watermark W_{cop} and expect the RecMark to recover $\hat{W}_{syn}$ and $\hat{W}_{cop}$ after passing through the differentiable Distortion Layer. In the inference process, the original video V_{ori} is partitioned into groups of 8 frames, watermarks are embedded every other group using two pre-trained encoders. During extraction, the group of frames that can correctly reconstruct the synchronization watermark is aligned and then fed into CopDecoder to retrieve the copyright watermark.

learning under asynchronous recording conditions.

- To mitigate the performance degradation caused by temporal misalignment, we develop a learned frame synchronization mechanism implemented through a dual-watermark framework. A dedicated synchronization watermark is embedded in the frequency domain to provide frame-level synchronization cues and reconstruct temporally aligned frame groups, while the copyright watermark is embedded in the spatiotemporal domain through an encoder equipped with a Spatiotemporal Fusion Module (SFM).
- Extensive experiments demonstrate that RecMark achieves strong watermark robustness under camera recording attacks while maintaining high perceptual video quality. Ablation studies further verify the effectiveness of the temporal distortion modeling and SFM.

II. METHOD

As illustrated in Fig. 1, we introduce a differentiable distortion layer to simulate camera recording-induced temporal distortion, which enables the model to learn robust representations under asynchronous conditions. To address the resulting temporal misalignment during extraction, we further design a synchronization branch consisting of a SynEncoder and SynDecoder, which embeds and recovers frame-level synchronization watermarks for reliable frame alignment. On this basis, a CopEncoder and CopDecoder are used to embed and extract copyright information from temporally aligned video frames across the entire sequence.

A. Watermark Embedding

SynEncoder: Frame-level synchronization is crucial for resisting camera recording attacks, as temporal misalignment causes watermarked frames to shift from their original positions, thereby leading to the failure of watermark extraction. To address this, the SynEncoder embeds the synchronization watermark W_{syn} into the wavelet domain of cover frames F_{cov} . The frames are first transformed via discrete wavelet

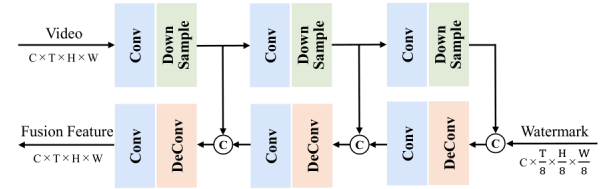


Fig. 2. Illustration of Spatiotemporal Fusion Module (SFM). The video and watermark are fed from opposite ends, downsampled and upsampled respectively, and progressively fused across multi-scales.

transform (DWT), and the mid- and low-frequency subbands are concatenated with the watermark and processed by a U-Net. The output is then transformed back to the spatial domain via inverse discrete wavelet transform (IDWT), generating the synchronized frames F_{syn} . During training, we randomly sample synchronization watermark to train the network. During inference, a fixed all-ones bitstring is embedded.

CopEncoder: As F_{syn} already carries synchronization information, we embed the copyright watermark W_{cop} in the spatiotemporal domain to avoid interference. To match video characteristics, we introduce a Spatiotemporal Fusion Module (SFM) that converts each watermark bit into a redundant representation aligned with the video feature space. This redundancy ensures that the watermark can still be reliably recovered even under partial corruption [20]. As illustrated in Fig. 2, SFM projects the watermark into multi-scale feature tensors via deconvolution and fuses them with video features extracted at different resolutions. At each scale, the watermark representation is concatenated with corresponding video features and progressively integrated in a coarse-to-fine manner, enabling adaptive alignment and diffusion across the spatiotemporal domain. The fused representation is then concatenated with the video and fed into a 3D U-Net to produce the copyright-watermarked frames F_{wm} .

B. Distortion Layer

The distortion layer is specifically designed to enhance robustness in training process by simulating distortions from

camera recording. We decompose the distortions into spatial components caused by display optics and camera sensor imperfections and temporal components caused by sampling inconsistencies. This decomposition enables a physically grounded simulation of recording degradations.

During the camera recording process, the camera lens forms an optical image by focusing and projecting light signals from the screen onto the image sensor. The distortions introduced in this stage, which primarily originate from the complex interplay between the display and camera, including optical imaging and subsequent signal conversion, are summarized as spatial distortions [21], [22], [23], [24]. Fundamentally, these can be conceptualized as frame-by-frame pixel-level degradations. Specifically, contrast distortion is modeled using pixel-wise linear transformations to emulate the pronounced intensity fluctuations arising from reflected ambient light superimposed onto the original screen content, leading to amplified brightness differences. Brightness distortion simulates global luminance variations in video frames caused by changes in ambient illumination during the optical imaging stage of camera capture. Perspective distortion is simulated via homography-based transformations, capturing geometric deformations induced by non-parallel alignment between the camera lens and the display, effectively mapping frames from one viewpoint to another. Additive Gaussian noise is introduced to approximate inevitable pixel-level stochastic perturbations arising from signal conversion in the imaging pipeline. Resolution scaling is implemented using bilinear interpolation to mimic spatial resizing effects caused by mismatched pixel densities between the display and the camera, as well as variations in shooting distance. Finally, Moiré artifacts are synthesized by superimposing sinusoidal gratings, capturing the complex interference patterns induced by sampling discrepancies. To facilitate reproducibility, we provide the detailed settings of the spatial distortions used in Table 1. All input frames are normalized to $[0, 1]$, and the distorted outputs are clipped to the same range. At each training iteration, all spatial distortions are applied. For perspective distortion, the four corners are independently displaced within $[-rW, rW]$ and $[-rH, rH]$ along the horizontal and vertical directions, respectively, and the resulting homography is applied to each frame.

During the temporal sampling stage, a camera acquires discrete frames of a continuously evolving scene at a fixed recording frame rate via the shutter, producing a time-ordered video sequence. From a sampling-theoretic perspective, this process can be interpreted as a non-uniform resampling of

the original discrete frame sequence, for which we construct a differentiable formulation to capture the inherent temporal asynchrony. To model these temporal distortions, we focus on sampling inconsistencies caused by frame rate mismatches between playback and recording devices, as shown in Fig. 3. These discrepancies lead to temporal desynchronization, which we represent through three operations, namely under-sampling, over-sampling, and mixed-sampling. Formally, let f_p and f_r denote the playback and recording frame rates, respectively, and let F_k be the watermarked frames, k indexes the frames in watermarked videos, and k' indicates the corresponding frames in the distorted video. The distorted video V_{dis} can be expressed as:

$$V_{dis} = \begin{cases} \{F_{k'} \mid k' = \lfloor \frac{k f_p}{f_r} \rfloor\}, & f_r < f_p \\ \{F_{k'} \mid k' = \lceil \frac{k f_p}{f_r} \rceil\}, & f_r > f_p \end{cases} \quad (1)$$

Here, when $f_r < f_p$, some frames are dropped to simulate under-sampling, while $f_r > f_p$ results in repeated frames to emulate over-sampling. In addition, asynchronous recording and playback start times, as well as frame-rate mismatches, may cause the camera exposure interval to span the transition between consecutive playback frames, resulting in a captured frame that contains mixed information from multiple playback frames. Mixed-sampling simulates chaotic frames overlapping by linearly combining adjacent frames, capturing inter-frame distortions caused by sampling inconsistencies:

$$V_{dis} = \{\alpha_k F_k + (1 - \alpha_k) F_{k+1}\}, \quad \alpha_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(0, 1). \quad (2)$$

Finally, a pretrained 3D U-Net-based compression network is incorporated into the distortion layer to simulate video compression and enhance robustness against video encoding. During training, all spatial distortions are applied, whereas a single temporal distortion is randomly selected with equal probability at each step.

C. Watermark Extraction

SynDecoder: During training, SynDecoder receives frames after distortion and extracts the synchronization watermark $\hat{W}_{syn}$ via DWT and U-Net. During inference, it is used to recover synchronization watermark from the distorted video V_{dis} and each synchronization bit is binarized with a threshold of 0.5. Through an alignment operation, frames where extraction fails are removed, and the successfully extracted sequence of frames forms a group that is then fed into CopDecoder. Unlike classifier-based frame identification, which is sensitive

TABLE I
DETAILED PARAMETER SETTINGS OF THE SPATIAL DISTORTIONS.

Distortion	Formulation	Parameter setting
Contrast	$\mathbf{F}' = c(\mathbf{F} - \mu) + \mu$	$c \sim \mathcal{U}(0.95, 1.05)$
Brightness	$\mathbf{F}' = b\mathbf{F}$	$b \sim \mathcal{U}(0.8, 1.2)$
Perspective	$\mathbf{F}' = \mathcal{H}(\mathbf{F}; \Delta\mathbf{P})$	Maximum corner displacement: $0.1 \times (H, W)$
Gaussian noise	$\mathbf{F}' = \mathbf{F} + \epsilon$	$\epsilon \sim \mathcal{N}(0, 0.01^2)$
Scaling	$\mathbf{F}' = \text{Resize}(\text{Resize}(\mathbf{F}, sH, sW), H, W)$	$s \sim \mathcal{U}(0.8, 1.2)$, Resize = bilinear interpolation
Moiré pattern	$\mathbf{F}' = \mathbf{F} + \mathbf{M}$, $\mathbf{M}(x, y) = A \sin[2\pi f(x \cos \theta + y \sin \theta) + \phi]$	$A \sim \mathcal{U}(0.01, 0.05)$, $f \sim \mathcal{U}(0.03125, 0.125)$, $\theta \sim \mathcal{U}(0, \pi)$, $\phi \sim \mathcal{U}(0, 2\pi)$

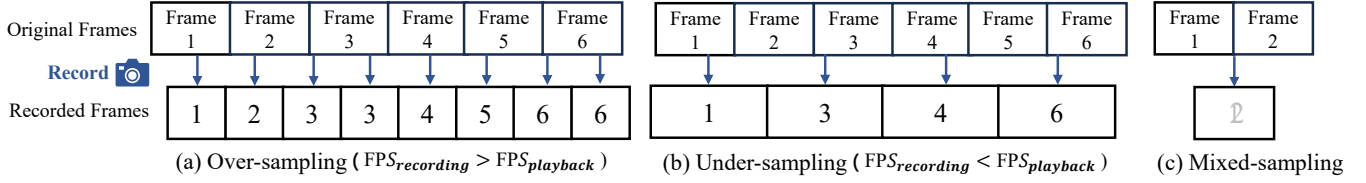


Fig. 3. Illustration of three temporal distortions. When a camera records a video played on a screen, frame rate mismatches can cause over-sampling and under-sampling. Meanwhile, when the sampling coincides with frame transition, mixed-sampling occurs overlapping.

to camera recording attacks, the SynDecoder robustly aligns watermarked frames through training with the distortion layer.

CopDecoder: We train CopDecoder to extract the copyright watermark $\hat{W}_{cop}$ from an 8-frame group by aggregating multi-scale features. Hierarchical features are obtained via downsampling and convolutional branches, concatenated, and fed into fully connected layers to recover the 96-bit watermark.

D. Training

The training process of RecMark can be divided into two stages. Given the 8 cover frames F_{cov} and a 96-bit copyright watermark W_{cop} , we first train CopEncoder and CopDecoder using a combination of binary cross-entropy loss (BCE) for the watermark extraction and mean squared error loss (MSE) for visual fidelity:

$$\mathcal{L}_{cop} = BCE(W_{cop}, \hat{W}_{cop}) + \lambda MSE(F_{cov}, F_{wm}) \quad (3)$$

where λ balances the visual MSE loss and is set to 3. Furthermore, we freeze the weights of CopEncoder and CopDecoder and jointly train the SynEncoder and SynDecoder, the loss is:

$$\mathcal{L}_{syn} = BCE(W_{syn}, \hat{W}_{syn}) + \gamma MSE(F_{cov}, F_{wm}) \quad (4)$$

where γ balances the visual MSE loss and is set to 5.

III. EXPERIMENTS

A. Experimental Setup

- **Dataset:** We train RecMark on the Vimeo-90K [25] and evaluate it on 50 testing videos sampled from different categories of Kinetics-400 [26]. During training, all frames are resized to 128×128 , and each sample consists of 8 consecutive frames. For video clips containing fewer than 8 frames, the missing frames are randomly padded. During testing, 64 consecutive frames are selected from each video and resized to 128×128 . To validate the effectiveness of RecMark on real-world long videos, each video is divided into 8 groups of 8 frames, and the watermark is embedded into the even-numbered groups. Unless otherwise specified, a 96-bit payload is used in the main experiments.
- **Training Details:** We use the AdamW optimizer [27] to minimize the loss function with a learning rate of $1e-4$ in two training stages. Each stage is trained for 100 epochs with a batch size of 4. All experiments are executed on an NVIDIA RTX 3080 GPU.
- **Comparison:** To evaluate RecMark against representative learning-based watermarking approaches, we compare it with two state-of-the-art deep video watermarking methods, REVMARK [18] and VideoSeal [19]. We also include

two screen-shooting-resistant image watermarking methods, StegaStamp [21] and PIMoG [22], which are applied independently to individual video frames. In our experiments, only RecMark has the capability of automatic synchronization. Since only RecMark performs automatic temporal synchronization, frame indices were displayed outside the evaluated region to align the recorded frames for competing methods. The index region was removed before extraction and therefore did not affect watermark decoding. The screen we use for experiments is “ENVISION G249G”, and the mobile phone is “iPhone 15 Pro”. All compared methods are configured according to their original papers. Among them, the two video watermarking baselines use the same 96-bit payload as RecMark, while StegaStamp [21] and PIMoG [22] use payloads of 100 bits and 30 bits, respectively.

B. Watermark Robustness

To evaluate the robustness under different playback frame rates f_p and recording frame rates f_r , we conduct corresponding experiments. Videos are played on the screen at three common playback frame rates and recorded using two typical camera frame rates, with a 30 cm screen-to-camera distance. The results are presented in Table 2. We can see that the proposed RecMark consistently achieves high accuracy, exceeding 97% across all tested frame rate combinations, demonstrating its strong robustness. We also conducted experiments with the screen and recording device positioned at varying distances. Videos are played at a fixed frame rate of 25 fps while being recorded at 30 fps. The results are presented in Table 3. Despite a slight drop in accuracy at larger distances, our method consistently achieves the highest performance. At the same time, as shown in Table 4, RecMark demonstrates strong robustness against various digital distortions. For a fair comparison, the parameter settings of these distortions are identical to those adopted in REVMARK [18].

TABLE II
EXTRACTION ACCURACY WITH DIFFERENT PLAYBACK AND RECORDING FRAME RATES.

f_r (fps)	30			60		
f_p (fps)	25	30	60	25	30	60
REVMARK [18]	80.42%	83.06%	56.94%	77.85%	73.58%	82.31%
VideoSeal [19]	50.18%	49.92%	50.27%	51.03%	49.61%	50.84%
StegaStamp [21]	93.12%	95.08%	87.96%	92.84%	92.31%	94.02%
PIMoG [22]	94.08%	94.57%	85.96%	90.76%	91.42%	94.21%
Proposed	98.72%	99.06%	98.21%	97.96%	98.61%	99.74%

TABLE III
EXTRACTION ACCURACY WITH DIFFERENT SHOOTING DISTANCES.

Distance (cm)	30	40	50
REVMARK [18]	80.42%	82.46%	81.87%
VideoSeal [19]	50.18%	50.18%	50.62%
StegaStamp [21]	93.12%	93.72%	91.35%
PIMoG [22]	94.08%	91.42%	92.16%
Proposed	98.72%	97.71%	96.61%

TABLE IV
EXTRACTION ACCURACY UNDER DIGITAL DISTORTIONS.

Distortion	Gaussian Noise ($\sigma=0.04$)	Gaussian Blur ($\sigma=2.0$)	Random Scaling ($p=0.4$)	H.264/AVC (CRF=22)
REVMARK [18]	99.94%	99.76%	84.21%	95.52%
VideoSeal [19]	99.97%	93.42%	91.23%	94.35%
StegaStamp [21]	99.79%	97.65%	95.91%	94.21%
PIMoG [22]	99.55%	88.37%	91.54%	93.61%
Proposed	99.96%	98.61%	98.37%	98.72%

In the camera recording experiments, video watermarking baselines fail to extract watermarks due to frame rate mismatches, while image watermarking baselines also suffer reduced accuracy under real camera recording attacks. In conclusion, our method exhibits strong robustness, confirming the effectiveness of our distortion analysis and network design.

C. Device Generalization and Run-to-Run Stability

To evaluate the generalization of RecMark across different playback and recording devices, we conducted camera recording experiments using two displays and two smartphones. The playback devices included an ENVISION G249G external monitor and the built-in display of a Lenovo Legion Y9000P laptop, while the recording devices included an iPhone 15 Pro and an OPPO Reno 13. All four display-camera combinations were evaluated under the same protocol: the recording distance was fixed at 30 cm, the video playback frame rate was set to 25 FPS, and the camera recording frame rate was set to 30 FPS. The results are reported in Table 5. RecMark maintains consistently high extraction accuracy across the four device combinations, demonstrating its robustness to variations in display and recording devices under the evaluated conditions.

TABLE V
EXTRACTION ACCURACY UNDER DIFFERENT DEVICE COMBINATIONS.

Device	ENVISION G249G	Lenovo Legion Y9000P laptop
iPhone 15 Pro	98.72%	98.44%
OPPO Reno 13	97.50%	98.65%

To evaluate run-to-run variability, we independently repeated the camera recording experiment five times using the ENVISION G249G display and the iPhone 15 Pro. Before each trial, video playback and camera recording were restarted independently, while the recording distance, playback frame rate, and recording frame rate were fixed at 30 cm, 25 FPS, and 30 FPS, respectively. The complete set of 50 test videos was evaluated in each trial. Across the five independent recordings, RecMark achieved a mean extraction accuracy of 99.03%, with a sample standard deviation of 0.63 percentage points.

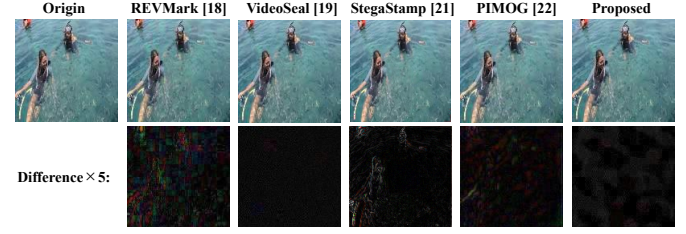


Fig. 4. Visual examples of watermarked frames and the residual difference multiplied by 5.

The limited run-to-run variation demonstrates the stability of RecMark under the evaluated physical recording condition.

D. Video Quality

To quantitatively evaluate imperceptibility, we measure the average PSNR and SSIM of the watermarked videos. Table 6 reports video quality across methods. Image watermarking baselines show lower visual quality due to redundant per-frame embedding, while video watermarking methods achieve better perceptual quality. Our approach further reduces degradation, with negligible impact on video quality. Although VideoSeal [19] achieves comparable quality, it fails to extract watermarks in the previously described robustness tests. Fig. 4 illustrates example frames, where our method yields the most imperceptible residuals.

TABLE VI
THE PSNR AND SSIM VALUES FOR EACH METHOD.

Method	REVMARK [18]	VideoSeal [19]	StegaStamp [21]	PIMOG [22]	Proposed
PSNR (dB)	36.03	38.21	29.74	36.25	38.31
SSIM	0.984	0.987	0.984	0.979	0.987

E. Synchronization Watermark Analysis

To directly evaluate the proposed synchronization mechanism, we visualize the frame-level predictions of SynDecoder and compare them with the ground-truth synchronization labels. Each test video contains 64 frames divided into eight groups, with synchronization and copyright watermarks embedded into the even-numbered groups. During camera recording, numerical frame indices are displayed outside the evaluated video region solely to establish the ground-truth temporal positions. For each recorded frame, SynDecoder produces a synchronization probability p_t . Fig. 5 compares the predicted probabilities and ground-truth labels.

As shown in Fig. 5, the synchronization probabilities predicted by SynDecoder closely match the ground-truth distribution of watermarked frames. High probabilities are consistently observed within the watermarked frame groups, whereas the predictions for non-watermarked groups remain close to zero. Despite minor probability fluctuations near the group boundaries caused by camera-recording distortions, the predictions remain well separated by the decision threshold of 0.5. These results provide direct qualitative evidence that the synchronization watermark remains detectable after camera recording. SynDecoder can accurately distinguish valid watermarked frames from non-watermarked or temporally

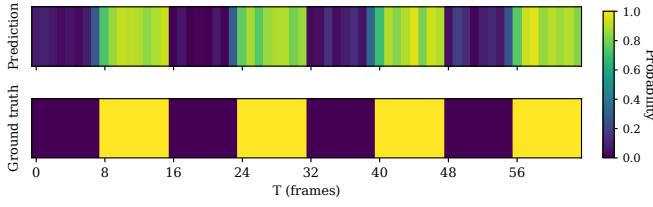


Fig. 5. Visualization of frame-level synchronization watermark detection under camera recording. The top row shows the synchronization probabilities predicted by SynDecoder, while the bottom row shows the corresponding ground-truth labels. A ground-truth label of 1 indicates a frame belonging to a watermarked group, whereas 0 indicates a non-watermarked frame. Predictions above 0.5 are regarded as successful synchronization detections.

misaligned frames, thereby enabling the alignment module to discard invalid frames and reconstruct the fixed-length frame groups required by CopDecoder. This agreement between the frame-level predictions and ground-truth labels demonstrates the effectiveness of the proposed synchronization mechanism under the evaluated recording condition.

F. Payload Analysis

To investigate the trade-off among payload capacity, perceptual quality, and robustness, we evaluated RecMark with copyright payloads of 30, 96, and 128 bits. The synchronization watermark configuration was kept unchanged across all settings. Camera-recording experiments were conducted at a fixed shooting distance of 30 cm. PSNR and SSIM were calculated between the original and watermarked videos before recording, while bit accuracy was measured after camera recording. The results are reported in Table 7.

TABLE VII

PERFORMANCE OF RECMark UNDER DIFFERENT COPYRIGHT PAYLOADS.

Payload (bits)	PSNR (dB)	SSIM	$f_r=30, f_p=60$	$f_r=60, f_p=30$
30	40.82	0.990	98.78%	98.89%
96	38.31	0.987	98.21%	98.61%
128	38.29	0.986	98.09%	97.98%

As the payload increases, PSNR, SSIM, and extraction accuracy decrease slightly, reflecting the expected trade-off between embedding capacity, perceptual quality, and robustness. Nevertheless, RecMark maintains satisfactory visual quality and extraction accuracy at 128 bits. The 96-bit payload adopted in the main experiments therefore provides a balanced configuration between capacity and robustness.

G. Ablation Studies

We conduct ablation studies to systematically evaluate the individual contributions of the SFM and the temporal distortion modeling. Specifically, three variants are considered: Full, No-SFM, and No-Temporal Distortion (No-TD). The quantitative results are summarized in Table 8. As observed, removing the SFM leads to noticeable degradation in both robustness and perceptual quality, indicating its critical role in stabilizing watermark embedding and preserving visual fidelity. In contrast, excluding temporal distortion modeling significantly weakens the method's resilience against camera recording scenarios, highlighting the importance of explicitly simulating temporal desynchronization effects during training.

The Full model achieves a balance between robustness and visual quality, confirming the effectiveness of both components.

TABLE VIII
ABLATION RESULTS OF RECMark.

Method	PSNR(dB)	SSIM	$f_r=30, f_p=60$	$f_r=60, f_p=30$
No-SFM	34.69	0.941	83.57%	84.31%
No-TD	42.21	0.996	65.12%	68.17%
Full	38.31	0.987	98.21%	98.61%

H. Computational Complexity Analysis

We further evaluate the computational complexity of RecMark in terms of floating-point operations (FLOPs) and the number of model parameters. In particular, StegaStamp and PIMoG process video frames independently; therefore, their computational cost is incurred repeatedly for every frame and they do not provide temporal modeling or automatic synchronization. For a fair comparison, FLOPs are normalized to the computational cost of processing one 8-frame group. For frame-wise image watermarking methods, the per-frame FLOPs are therefore multiplied by eight. As shown in Table 9, RecMark's computational cost is higher than that of some lightweight baselines. The relatively high FLOPs of RecMark mainly arise from its dual-watermark architecture, in which both synchronization and copyright watermark branches must be executed. Moreover, the 3D convolutional operations and the Spatiotemporal Fusion Module jointly process eight frames to capture temporal dependencies, while the DWT/IDWT-based synchronization branch introduces additional computation. This overhead supports automatic frame alignment and robust copyright extraction, which are not provided by the lightweight frame-wise baselines.

TABLE IX

COMPARISON OF FLOATING-POINT OPERATIONS (FLOPs) AND MODEL PARAMETERS.

Method	REVMARK [18]	VideoSeal [19]	StegaStamp [21]	PIMoG [22]	Proposed
FLOPs (G)	242	45	12	16	303
Param. (M)	8.4	40.1	2.7	3.5	15.7

IV. CONCLUSION

This paper presents RecMark, a dual-watermark framework that robustly protects video copyright against camera recording attacks. By incorporating a differentiable spatiotemporal distortion layer and synchronization-based temporal alignment, RecMark achieves high extraction accuracy while preserving video quality. Experiments demonstrate the effectiveness of RecMark across multiple scenarios. Nevertheless, rolling-shutter effects and motion blur remain unmodeled and will be addressed through more comprehensive acquisition modeling in future work.

REFERENCES

- [1] D. Wu, X. Zhang, J. Wang, L. Li, and G. Feng, "Novel robust video watermarking scheme based on concentric ring subband and visual cryptography with piecewise linear chaotic mapping," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 10, pp. 10281–10298, 2024.

- [2] M. Asikuzzaman and M. R. Pickering, "An overview of digital video watermarking," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 28, no. 9, pp. 2131–2153, 2017.
- [3] S. Chen, A. Malik, X. Zhang, G. Feng, and H. Wu, "A fast method for robust video watermarking based on zernike moments," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 12, pp. 7342–7353, 2023.
- [4] F. Zhang, H. Wang, M. He, L. Yang, and J. Li, "Adaptive video watermarking with perceptual guarantee and efficiency optimization," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 4610–4614.
- [5] M. He, H. Wang, F. Zhang, S. M. Abdullahi, and L. Yang, "Robust blind video watermarking against geometric deformations and online video sharing platform processing," *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 6, pp. 4702–4718, 2022.
- [6] H. Do, D. Choi, T. Kim, and H. Choi, "Digital video watermarking based on histogram and temporal modulation and robust to camcorder recording," in *Proceedings of the IEEE International Symposium on Signal Processing and Information Technology*, 2008, pp. 330–335.
- [7] P. A. Hernandez-Avalos, C. Feregrino-Urbe, R. Cumplido, and J. J. Garcia-Hernandez, "Towards the construction of a benchmark for video watermarking systems: temporal desynchronization attacks," in *Proceedings of the IEEE International Midwest Symposium on Circuits and Systems*, 2010, pp. 628–631.
- [8] P. Schaber, S. Kopf, S. Wetzel, T. Ballast, C. Wesch, and W. Effelsberg, "Cammark: Analyzing, modeling, and simulating artifacts in camcorder copies," *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 11, no. 2s, pp. 1–23, 2015.
- [9] L. Chen, C. Wang, X. Zhou, and Z. Qin, "Robust and compatible video watermarking via spatio-temporal enhancement and multiscale pyramid attention," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 2, pp. 1548–1561, 2025.
- [10] C. Hui, S. Liu, W. Shi, F. Jiang, and D. Zhao, "Spatio-temporal context based adaptive camcorder recording watermarking," *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 18, no. 3s, pp. 1–25, 2022.
- [11] Q. Chang, L. Huang, S. Liu, H. Liu, T. Yang, and Y. Wang, "Blind robust video watermarking based on adaptive region selection and channel reference," in *Proceedings of the ACM International Conference on Multimedia*, 2022, pp. 2344–2350.
- [12] L. Lin, D. Wu, J. Wang, Y. Chen, X. Zhang, and H. Wu, "Automatic, robust and blind video watermarking resisting camera recording," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 12, pp. 13413–13426, 2024.
- [13] H. Wang, H. Wang, M. He, F. Zhang, and J. Xia, "Robust video watermarking against digital editing and camcording," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 11, pp. 11068–11082, 2025.
- [14] K. A. Zhang, L. Xu, A. Cuesta-Infante, and K. Veeramachaneni, "Robust invisible video watermarking with attention," *arXiv preprint arXiv:1909.01285*, 2019.
- [15] X. Zhang, Y. Xu, R. Li, J. Yu, W. Li, Z. Xu, and J. Zhang, "V2a-mark: Versatile deep visual-audio watermarking for manipulation localization and copyright protection," in *Proceedings of the ACM International Conference on Multimedia*, 2024, pp. 9818–9827.
- [16] Z. Su, X. Qiu, H. Xu, T. Jiang, J. Zhuang, C. Yuan, M. Li, S. He, and F. R. Yu, "Safe-sora: Safe text-to-video generation via graphical watermarking," *arXiv preprint arXiv:2505.12667*, 2025.
- [17] X. Luo, Y. Li, H. Chang, C. Liu, P. Milanfar, and F. Yang, "Dvmark: a deep multiscale framework for video watermarking," *IEEE Transactions on Image Processing*, vol. 34, pp. 4371–4385, 2025.
- [18] Y. Zhang, J. Ni, W. Su, and X. Liao, "A novel deep video watermarking framework with enhanced robustness to h.264/avc compression," in *Proceedings of the ACM International Conference on Multimedia*, 2023, pp. 8095–8104.
- [19] P. Fernandez, H. Elshar, I. Z. Yalniz, and A. Mourachko, "Video seal: Open and efficient video watermarking," *arXiv preprint arXiv:2412.09492*, 2024.
- [20] Z. Jia, H. Fang, and W. Zhang, "Mbrs: Enhancing robustness of dnn-based watermarking by mini-batch of real and simulated jpeg compression," in *Proceedings of the ACM International Conference on Multimedia*, 2021, pp. 41–49.
- [21] M. Tancik, B. Mildenhall, and R. Ng, "Stegastamp: Invisible hyperlinks in physical photographs," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 2117–2126.
- [22] H. Fang, Z. Jia, Z. Ma, E.-C. Chang, and W. Zhang, "Pimog: An effective screen-shooting noise-layer simulation for deep-learning-based watermarking network," in *Proceedings of the ACM International Conference on Multimedia*, 2022, pp. 2267–2275.
- [23] L. Fu, X. Liao, J. Guo, L. Dong, and Z. Qin, "Waverecovery: Screen-shooting watermarking based on wavelet and recovery," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 4, pp. 3603–3618, 2024.
- [24] Y. Li, X. Liao, and X. Wu, "Screen-shooting resistant watermarking with grayscale deviation simulation," *IEEE Transactions on Multimedia*, vol. 26, pp. 10908–10923, 2024.
- [25] F. Perazzi, J. Pont-Tuset, B. McWilliams, L. Van Gool, M. Gross, and A. Sorkine-Hornung, "A benchmark dataset and evaluation methodology for video object segmentation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 724–732.
- [26] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev *et al.*, "The kinetics human action video dataset," *arXiv preprint arXiv:1705.06950*, 2017.
- [27] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017.



Zhengdong Wang received the M.S. degree from the College of Cyber Science and Technology, Hunan University. His research interests include the digital watermarking.



Xin Liao (Senior Member, IEEE) received the B.E. and Ph.D. degrees in information security from the Beijing University of Posts and Telecommunications in 2007 and 2012, respectively. He is currently a Professor with the College of Cyber Science and Technology, Hunan University, where he also serves as the Vice Dean. He worked as a Post-Doctoral Fellow with the Institute of Software, Chinese Academy of Sciences, and also a Research Associate with The University of Hong Kong. From 2016 to 2017, he was a Visiting Scholar with the University of

Maryland, College Park, USA. His current research interests include multimedia forensics, AI security, and watermarking. He is the Vice Chair of Technical Committee (TC) on Multimedia Security and Forensics of Asia-Pacific Signal and Information Processing Association, a member of TC on Computer Forensics of the Chinese Institute of Electronics, and a member of TC on Digital Forensics and Security of the China Society of Image and Graphics. He is serving as an Associate Editor for the IEEE Signal Processing Magazine.



Yufeng Wu received the M.S. degree from the School of Computer, Nanjing University of Information Science and Technology, Nanjing, China, in 2024. He is currently pursuing the Ph.D. degree with the College of Cyber Science and Technology, Hunan University. His research interests include the digital watermarking and multimedia security.